Finding the best feasible bundle

Lecture notes

What this lecture is about. Last lecture ended with a picture: the best feasible bundle is where the highest reachable indifference curve touches the budget line, and there $MRS = p_x/p_y$. Today we find that bundle with numbers, in four ways that all say the same thing: the picture, "bang for the buck", arrows, and the Lagrangian. Then we look at the cases where the rule does not apply (corners and kinks), and we practise.

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1 One example for the whole lecture

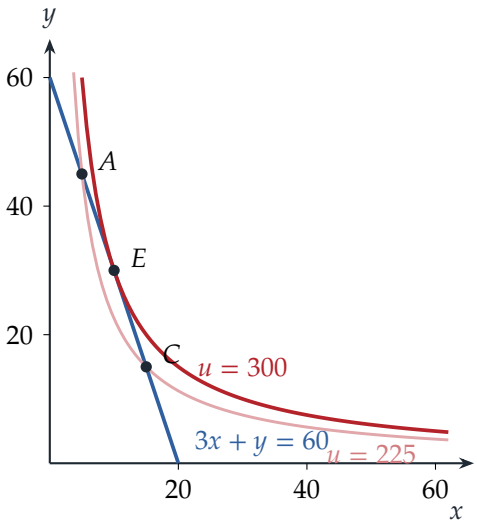
Example: the running example (the same as at the end of lecture 8)

Utility $u = x\,y$; prices $p_x = 3$, $p_y = 1$; money $B = 60$. The budget line is $3x + y = 60$. Three bundles on it:

$$A = (5, 45), \qquad C = (15, 15), \qquad E = (10, 30).$$

Check: $3 \cdot 5 + 45 = 3 \cdot 15 + 15 = 3 \cdot 10 + 30 = 60$. Which one is the best feasible bundle?

2 Way 1: the picture



A and C are equally good: $u(A) = 5 \cdot 45 = 225 = 15 \cdot 15 = u(C)$, so they lie on the same indifference curve, which *cuts* the budget line. The steepness of a curve at a bundle is its MRS,

here $MRS = y/x$ (we compute it in §3). At A it is $45/5 = 9$, steeper than the line's 3: moving along the line toward more x reaches better curves. At C it is $15/15 = 1$, flatter than the line: moving back toward more y helps. At E it is $30/10 = 3$, exactly the line's slope: the curve $u = 300$ just touches the line, and no better curve is reachable. E is the best feasible bundle.

3 Way 2: bang for the buck

Marginal utility

MU_x is how much u rises with one more unit of x, holding y fixed (a derivative). For $u = xy$: $MU_x = y$ and $MU_y = x$, so $MRS = MU_x/MU_y = y/x$. Check at A: from $(5, 45)$ to $(6, 45)$, u goes from 225 to 270, up by $45 = y$.

Divide each marginal utility by its price: utility per euro.

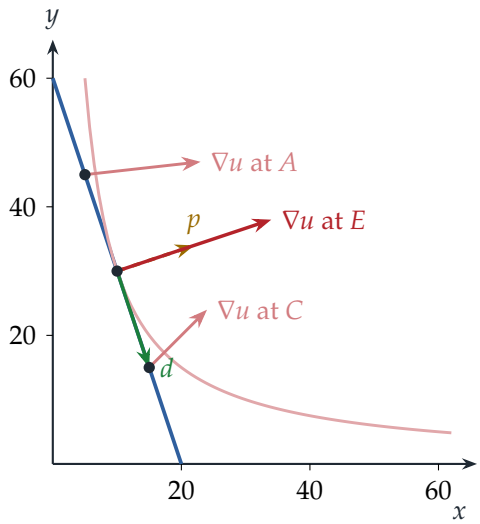
bundle	MU_x/p_x	MU_y/p_y	a euro moved ...
$A = (5, 45)$	$45/3 = 15$	$5/1 = 5$	from y to x gains
$C = (15, 15)$	$15/3 = 5$	$15/1 = 15$	from x to y gains
$E = (10, 30)$	$30/3 = 10$	$10/1 = 10$	gains nothing

At the best feasible bundle, the last euro does equally well on both goods. Keep the number 10 in mind.

4 Way 3: arrows

Dot product

For two arrows (a, b) and (c, d): $(a, b) \cdot (c, d) = ac + bd$. It is 0 exactly when the two arrows are perpendicular; it is positive when they point roughly the same way.



- **Along the line.** One more x and three less y cost the same: the step $d = (1, -3) = (p_y, -p_x)$ keeps you on the budget line, and so does any multiple of it, big or small.
- **The prices.** The price arrow $p = (3, 1)$ is perpendicular to the line: $p \cdot d = 3 \cdot 1 + 1 \cdot (-3) = 0$.
- **Better.** The arrow of marginal utilities points toward better bundles.

Gradient

The gradient $\nabla u = (MU_x, MU_y)$ is the arrow made of the two marginal utilities. It points in the direction in which utility rises fastest. It is perpendicular to the indifference curve: along the curve u does not change, so $\nabla u \cdot$ (a small step along the curve) $= 0$.

Which way along the line is better? Take the dot product of the gradient with the step d; its sign tells which way along the line is better (it is the projection of the gradient onto the line, up to the positive factor $|d|$):

bundle	∇u	$\nabla u \cdot d$	along the line
$A = (5, 45)$	$(45, 5)$	$45 - 15 = 30$	toward more x is better
$C = (15, 15)$	$(15, 15)$	$15 - 45 = -30$	toward more y is better
$E = (10, 30)$	$(30, 10)$	$30 - 30 = 0$	no step helps

In the picture: at A the gradient tilts toward more x, at C toward more y, and at E it lines up with p. At E the gradient has nothing left along the line: it is perpendicular to the line, like p. Two arrows perpendicular to the same line are multiples of each other (both point up and to the right, so the multiple is positive):

$$\nabla u = \lambda\, p, \qquad (30, 10) = 10 \cdot (3, 1).$$

Insight: three pictures, one condition

$\nabla u \cdot d = MU_x\, p_y - MU_y\, p_x = 0$ is the same as $\dfrac{MU_x}{MU_y} = \dfrac{p_x}{p_y}$ (MRS = price ratio, way 1) and as $\dfrac{MU_x}{p_x} = \dfrac{MU_y}{p_y}$ (equal bang for the buck, way 2).

5 Way 4: the Lagrangian

λ is a new unknown, solved for together with x and y; we will see that it is the utility value of one euro.

Lagrangian

$\mathcal{L} = u(x,y) + \lambda\,(B - p_x x - p_y y)$: utility, plus λ times the money left over. Setting its derivatives to zero gives the arrow condition and the budget line at once.

For our example, $\mathcal{L} = x\,y + \lambda\,(60 - 3x - y)$, and

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x} &= y - 3\lambda = 0, \\ \frac{\partial \mathcal{L}}{\partial y} &= x - \lambda = 0, \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= 60 - 3x - y = 0. \end{aligned}$$

The first two say $(y, x) = \lambda\,(3, 1)$: the gradient is a multiple of the price arrow (way 3). The third is the budget line. Solve: $x = \lambda$, $y = 3\lambda$, so $y = 3x$ (MRS $y/x = 3$ = price ratio); then $3x + 3x = 60$ gives $x = 10$, $y = 30$, and $\lambda = 10$, the bang for the buck at E.

The Lagrange multiplier λ

The number with $\nabla u = \lambda\, p$ at the best feasible bundle. It measures how much utility one more euro of money buys.

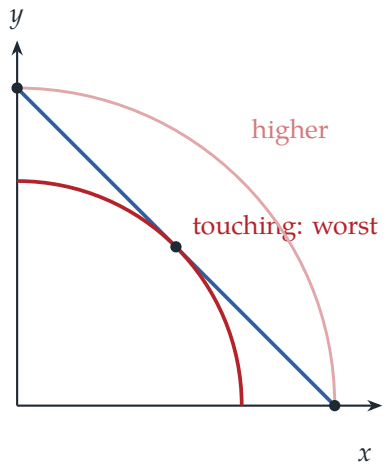
With money B the best feasible bundle is $x = B/6$, $y = B/2$, so $u^* = B^2/12$. Give more money in smaller and smaller steps:

B	best feasible bundle	u^*	$\Delta u^*/\Delta B$
60	$(10, 30)$	300	
61	$(10.167, 30.5)$	310.083	10.083
60.1	$(10.017, 30.05)$	301.0008	10.008
60.01	$(10.0017, 30.005)$	300.1000	10.0008

The ratio settles at $\lambda = 10$. (The number depends on how we number utility: with $u = 2xy$, λ doubles to 20. Its meaning, utility per extra euro at the best feasible bundle, does not.)

6 When the rule does not apply

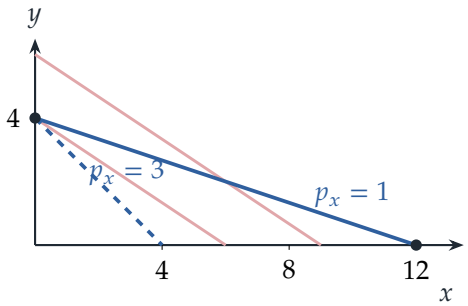
The rule finds a best feasible bundle where an indifference curve *touches* the line away from its ends. It works because the curves bow in. On a curve that bowed out, the touching point would be the *worst* bundle on the line: the ends of the line lie on higher curves.



Two exceptions to the rule:

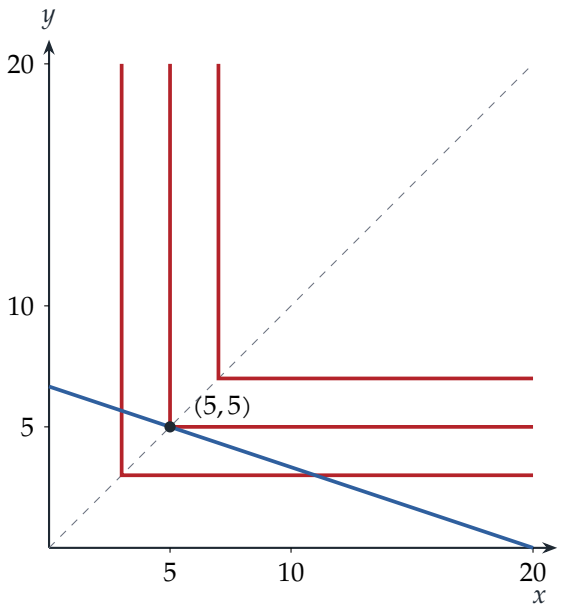
- **Corners.** If a step along the line always helps in one direction, you keep going until the line ends: all money on one good. There the gradient is not a multiple of p, yet the corner is the best feasible bundle, because the only improving step points off the end of the line.
- **Kinks.** If the indifference curve has a corner, its slope (the MRS) is not defined there.

Perfect substitutes: corners. $u = 2x + 3y$, so $MRS = 2/3$ everywhere; $p_y = 3$, $B = 12$.



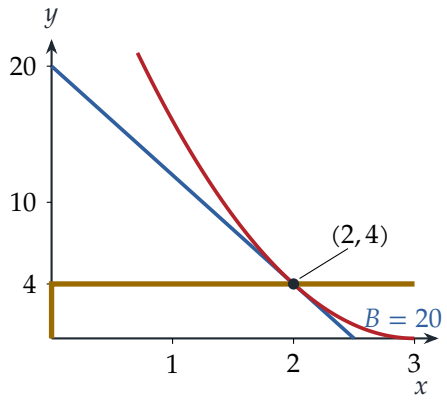
With $p_x = 1$ the price ratio is $1/3 < 2/3$: every euro does more on x (2 per euro against 1), so all on x, the corner $(12, 0)$. With $p_x = 3$ it is $1 > 2/3$: all on y, $(0, 4)$. With $p_x = 2$ the price ratio equals the MRS, the indifference line lies on the budget line, and every point on it is best. In arrows: $\nabla u = (2, 3)$ is a multiple of $p = (p_x, 3)$ only when $p_x = 2$.

Perfect complements: the kink. $u = \min\{x, y\}$, $p_x = 1$, $p_y = 3$, $B = 20$.



Buy in the ratio $x = y$: anything unmatched is wasted money. On the line, $x + 3x = 20$, so the best feasible bundle is $(5, 5)$. At the kink there is no single slope, so "MRS = price ratio" cannot be used; the ratio rule replaces it.

Quasi-linear: interior or corner, depending on money. $u = 2x + \sqrt{y}$, so $MU_x = 2$, $MU_y = 1/(2\sqrt{y})$ and $MRS = 4\sqrt{y}$; take $p_x = 8$, $p_y = 1$.



In the interior, $4\sqrt{y} = 8$ gives $y = 4$, whatever the money, once there is at least €4. With $B = 20$: $x = (20 - 4)/8 = 2$, the bundle $(2, 4)$. With $B = 2$ there is not even enough for $y = 4$: all on y, the corner $(0, 2)$. As money grows, the best feasible bundle (the gold path) first climbs the y-axis to $y = 4$, then moves straight right: extra money goes only into x.

7 Worked exercises

Example: 1 · climbing and cinema (an interior optimum)

You spend $B = 72$ euros a week on climbing visits (x, €6) and cinema tickets (y, €12); $u = x^{1/3}y^{2/3}$.

- $MU_x = \frac{1}{3}x^{-2/3}y^{2/3}$, $MU_y = \frac{2}{3}x^{1/3}y^{-1/3}$, so

$$MRS = \frac{\frac{1}{3}x^{-2/3}y^{2/3}}{\frac{2}{3}x^{1/3}y^{-1/3}} = \frac{1}{2}\cdot\frac{y^{2/3+1/3}}{x^{1/3+2/3}} = \frac{y}{2x}.$$

- $\frac{y}{2x} = \frac{6}{12}$ gives $y = x$; the budget $6x + 12y = 72$ gives $(4, 4)$, with $u^* = 4$.
- Bang for the buck at $(4, 4)$: $\frac{1/3}{6} = \frac{2/3}{12} = \frac{1}{18}$. The Lagrangian gives the same bundle and $\lambda = \frac{1}{18}$.
- For any B, $x = y = B/18$ and $u^* = B/18$: here every extra euro adds exactly $\lambda = 1/18$.

Example: 2 · gel pens and pencils (a corner)

$u = 3x + 2y$ (x gel pens, y pencils), so $MRS = 3/2$ everywhere; budget €12, a pencil €1. With a gel pen at €2 the price ratio is $2 > 3/2$: all on pencils, $(0, 12)$. At €1.20 the ratio is $1.2 < 3/2$: all on gel pens, $(10, 0)$.

Example: 3 · toy tricycles (a kink)

Each tricycle needs 1 frame and 3 wheels: $u = \min\{3x, y\}$ (the same ranking as $\min\{x, y/3\}$, the number of tricycles). A frame costs €30, a wheel €10, budget €180. One tricycle costs €60, so the best feasible bundle is $(3, 9)$. At the kink there is no single slope, so "MRS = price ratio" cannot be used; anything off the ratio $y = 3x$ is wasted money.

Summary

- Inside the budget line, the best feasible bundle satisfies $MRS = p_x/p_y$; equivalently equal bang for the buck, or $\nabla u = \lambda p$.
- The Lagrangian turns this into algebra; λ is the utility one more euro buys at the best feasible bundle.
- Corners: compare the MRS with the price ratio and spend everything on one good. Kinks: buy in the fixed ratio.

Further reading. Frank, *Microeconomics and Behavior*, pp. 71–76 and Appendix 3A, pp. 90–92 (week 5 page, additional readings; the password is given in class).